

Instanton-Based Efficient Computation with Prefactor Adjustments for Extreme Fluctuations in Turbulent Flows

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Abstract

In this paper, we first introduce the historical significance of turbulent flows in Section 1. In Section 2, we introduce the Desniansky-Novikov shell model, a discretized dynamical model inspired by the Navier-Stokes equations that is better suited for numerical analysis. Direct numerical simulations (Monte Carlo simulations) naturally fit this environment to compute event probabilities. In Section 3, we show the derivation of the instanton calculation. We then extend this derivation in Section 4 to account for second-order fluctuations around the instanton. These fluctuations take the form of a prefactor. Finally, in Section 5, we draw conclusions about the three methodologies for extreme event probability computations and the utility of the Desniansky-Novikov shell model.

1 Introduction

Turbulent flows in fluid dynamics are a complex mechanism where coherent structures interact in time and space, on a vast array of spatial scales from atmospheric flows that span kilometers to pouring a glass of water. The core intricacy of turbulent flows, and a major challenge in its modeling, is that energy is most readily injected at large scales, while dissipation only occurs at scales many orders of magnitude smaller. In between the large scale energy injection range and the much smaller scale energy dissipation range, there is a wide range of scales where these structures exhibit patterns in their behavior and a power-law spectrum for energy that has already been well-recorded and studied [1].

To this end, shell models for turbulent energy cascade can present highly non-trivial and intermittent behavior highly analogous to fully developed turbulence. At the same

time, shell models are more numerically and analytically tractable than the Navier-Stokes equations, which describe the motion of fluid substances by balancing momentum with forces like pressure and viscosity.

In studying out-of-equilibrium fluid dynamics in both micro- and macroscopic scales, strongly non-Gaussian and intermittent fluctuations are observed frequently, implying that abrupt changes in fluid velocity and energy occur more often than would be if standard Gaussian statistics were followed. These fluctuations result from randomness in both injection (at large scales) and dissipation (at small scales) of energy.

The challenge that arises in determining the statistical properties of such intrinsically rare events is computing statistical descriptors efficiently. By using Shell Models to mimic turbulent energy cascades while maintaining an efficient environment to sample extreme events

numerically, we can more readily investigate different properties of extreme events and various observable metrics in classical hydrodynamic turbulence.

Instanton-based methodologies have been successfully examined for this very purpose, using the Desniansky-Novikov shell model, by Goedert and Biferale [2]. This same model can be extended beyond hydrodynamics turbulence, an initial testing ground, to further settings such as modeling magnetohydrodynamics, passive scalars, and convective turbulence.

As referenced in the conclusion of [2], instanton-based methodologies can be improved by taking into account second-order fluctuations around the instantons, calculated as a prefactor function that adjusts the saddle-point approximations of the instantons to better match the shape of real-world observed distributions, which is what we explore in this paper. We also expand this research into target observable metrics beyond velocity $\sum_n u_n$, to also include velocity gradient $\sum_n u_n k_n$ and energy dissipation rate $\nu \sum_n u_n^2 k_n^2$. Adjusting for second-order fluctuations introduces a tradeoff between the accuracy of the methodology and the marginal runtime required to compute the prefactor. This extended methodology can be further applied to additional modeling settings beyond hydrodynamics turbulence as well.

For exponentially rare events in fluid dynamics, from atmospheric dynamics and convective bursts in storms to ocean mixing and breaking-wave or internal-tide events, the efficient computation of extreme events in fluid dynamics can replace the prohibitive cost of brute-force sampling, particularly of the Navier-Stokes equations, with a single optimization problem per target event, yielding absolute probabilities rather than mere scaling estimates once second-order corrections are included.

2 The Desniansky-Novikov Shell Model

Shell models allow for closer inspection of the Navier-Stokes equations in Fourier space by preserving many of its nonlinear properties and symmetries while being more tractable in its parameterization. Shell models are formed by discretizing Fourier space into concentric shells

where each shell is represented by a wavenumber, its radius, and each wavenumber is determined as part of a geometric sequence ($k_n = k_0 h^n$). Then, a single or a few metrics can be collected from each shell which contributes to the further simplifications on the Navier-Stokes equations, allowing for a logarithmic lattice, which in turn significantly decreases the degrees of freedom required to simulate solutions.

The real-valued Desniansky-Novikov (D.N.) shell model with an additive stochastic forcing reads as:

$$\begin{aligned} \frac{du_n}{dt} &= G_n[u] + f_n \\ \langle f_n(t) f_m(t') \rangle &= \chi_{nm} \delta(t - t') \\ G_n[u] &= c_1(k_n u_{n-1}^2 - k_{n+1} u_n u_{n+1}) \\ &\quad + c_2(k_n u_{n-1} u_n - k_{n+1} u_n^2) \\ &\quad - \nu k_n^2 u_n \end{aligned} \quad (1)$$

where c_1 , c_2 , and ν are real, constant parameters, with $\nu \geq 0$ corresponding to the viscosity of the fluid. The $f_n(t)$ terms represent the injections of velocity (and thereby energy) into the system at each time step, and χ_{nm} represents the correlation between the forcing applied to different shells n and m . Note that we define $\chi = k^{-6}$ in the simulation, demonstrating the injection of energy primarily at larger scales and significantly less at smaller scales. Each $\nu k_n^2 u_n$ term represents energy dissipating out of the system, demonstrating the contrasting property that energy dissipates much more readily at smaller scales.

Importantly, although this is in theory an infinite-dimensional dynamical system, for computational efficiency, we restrict the shell space to $n \in \{0, 1, \dots, N\}$ while imposing boundary conditions $u_{-1} = u_{N+1} = 0$ and choosing N large enough to include integral, inertial, and viscous scales. The D.N. shell model in particular is designed to preserve energy $E = \frac{1}{2} \sum u_n^2$ in inviscid and unforced settings, which means zero energy dissipation and zero energy injection.

Another core property of this model shows the preservation of the power law spectrum across shells, analogous to three-dimensional isotropic turbulence. In settings with high viscosity, energy tends to dissipate before reaching

smaller scale shells. An example energy spectrum is shown below, with data collected at a time in which all simulations reached stationarity:

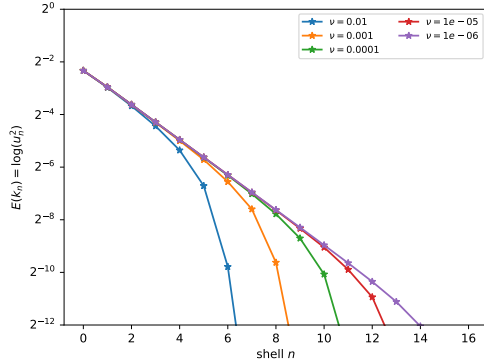


Fig. 1: Log Energy Spectrum with parameters $c_1 = 1.0$ and $c_2 = 0.0$

This is also known as a turbulent energy cascade, and it is another key similarity between shell models and fully developed turbulence. However, rare events are much easier to sample (though still very runtime expensive) in direct numerical simulations of shell models than in the Navier-Stokes equation, again highlighting the utility of simple shell models like this one at calculating the probability of extreme events. Especially the D.N. shell model, which parameterizes c_1 and c_2 , controls for the nonlinearity and asymptotic properties, allow for careful inspection and calculations to be made.

3 Instanton calculation

Strong non-Gaussian and intermittent fluctuations are observed in many out-of-equilibrium micro- and macroscopic fluids, implying that abrupt changes in fluid velocity occur more often and with larger effects than would be observed if Gaussian statistics were followed. Efficient modeling of the same large scale energy injection and small scale energy dissipation dynamical system as described by the Navier-Stokes equations can overcome the difficulty of observing extreme and important physical phenomena that arises from their intrinsic rarity. By leaning into the coherent structures that form the basis of such extreme phenomena, numerical tools are capable of quantitatively

assessing the statistical weight of such coherent spatio-temporal instanton-type solutions.

3.1 Path integral optimization formulation

As in [2], we base our path integral formulation on Martin-Siggia-Rose-Janssen-de Dominicis (MSRJD) path integral formalism. The general setup treats our differential equation as a field, so that the representative variable $u(x, t)$ is an n -dimensional vector (where we truncate the Fourier space to n discretized shells) that is subject to normal stochastic forcing f which is white in time and has some desired correlation χ ,

$$\begin{aligned} \dot{u} &= N[u] + f(t) \\ \langle f(t), f(t') \rangle &= \chi \delta(t - t') \end{aligned} \quad (2)$$

In particular, $u = [u_0, u_1, \dots]$, $N[u] = [G_0[u], G_1[u], \dots]$ and the stochastic forcing is also defined element-wise for the discrete shell space, $f(t) = [f_0, f_1, \dots]$, where $\langle f_n(t), f_m(t') \rangle = \chi_{n,m} \delta(t - t')$.

Because of this formulation, we can actually treat the randomness as a fixed parameter and find the most likely path to reach a certain extreme event; this is the overarching goal of instanton computations. Then we call the time steps of randomness η , i.e. defining $\eta = \frac{\Delta W}{\Delta t}$, since the stochastic forcing is modeled as standard Brownian motion. The resultant differential equation becomes

$$\dot{u} = G[u] + \sqrt{\chi} \eta - \nu k^2 u \quad (3)$$

where $G_n[u]$ is the standard D.N. coupling as defined previously. Then from this formulation, we can track the "cost" of randomness from a given realization of randomness through the rate function,

$$\begin{aligned} I(z) &= \frac{1}{2} \|\eta\|_{L^2}^2 \\ &= \frac{1}{2} \int_0^T \sum_{n=1}^N \eta_n^2(t) dt \end{aligned} \quad (4)$$

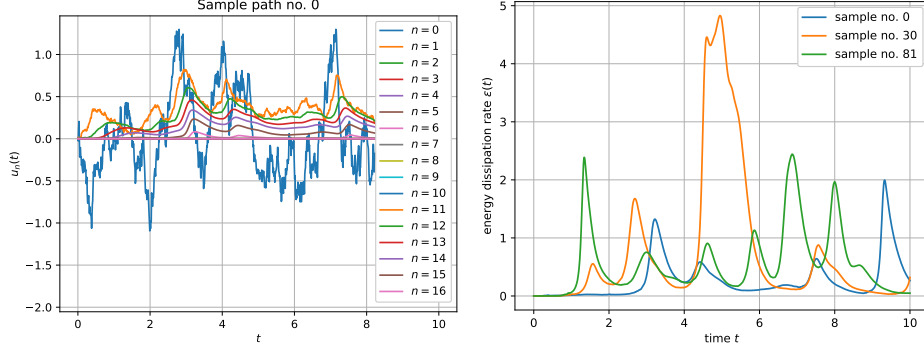


Fig. 2: Sample solution for the stochastically forced D.N. shell model (1) from the direct numerical simulation on the left and three sample time series of energy dissipation rates $\nu \sum_n u_n^2 k_n^2$ under the same parameters, $c_1 = 1.0$, $c_2 = 0.0$, $\nu = 0.01$, and $\chi_{nm} = k_n^{-6} \delta_{nm}$.

where z is a given target observable value for some predefined target metric, and the operation performed is the L^2 norm, which sums the randomness across all shells and across all time steps. If we take this to be like the "cost" of randomness, then intuitively, we expect the opposite of the rate function to be similar to the true PDFs of the observables, i.e. $\rho(z) \sim e^{-I(z)}$.

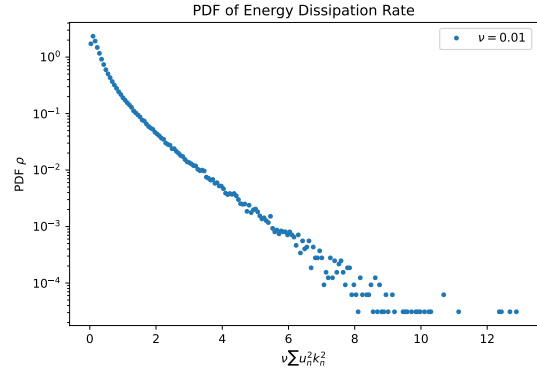
Then by finding the minimizing randomness realization, we would find the "cheapest" or least unlikely path, which therefore must be the highest likelihood path. So we define our optimization problem as parameterized by the randomness realization η ,

$$\min_{\eta} \frac{1}{2} \|\eta\|_{L^2}^2 \quad (5)$$

subject to the constraint $F(\eta) = z$ where $F(\eta)$ defines the target metric at a time of stationarity T , and z is the target observable value. We will notate this minimizing η as η_z throughout the remainder of the paper.

3.2 Solving the optimization problem

To solve this optimization problem, we first transform it from a constrained optimization problem to an unconstrained problem using the method of Lagrange multipliers, which solves the following objective function at each iteration,



Parameters used were $T = 120$, $\nu = 0.01$, $c_1 = 0.01$, and $c_2 = 0.0$

Fig. 3: Log plot of the histogram of energy dissipation rates $\epsilon = \nu \sum_n u_n^2 k_n^2$ from the direct numerical simulation when $c_1 = 1.0$, $c_2 = 0.0$, $\nu = 0.01$, and $\chi_{nm} = k_n^{-6} \delta_{nm}$. The average value is $\langle \epsilon \rangle = 0.508$.

$$\mathcal{L}(\eta, \lambda) = \frac{1}{2} \|\eta\|_{L^2}^2 - \lambda(F(\eta) - z) \quad (6)$$

where η and the Lagrange multiplier λ are updated after each iteration. The goal is to then look for a saddle point, which would minimize over η and maximize over λ . Then at the saddle, the two conditions should hold simultaneously:

$$\begin{aligned} \nabla_{\eta} \mathcal{L} &= 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= F(\eta) - z = 0 \end{aligned} \quad (7)$$

Note that the first condition is the optimality condition for η given the correct value of λ . The second condition just incorporates the constraint into the unconstrained equation.

However, two issues arise from this objective function. First, this formulation does not provide a principled update rule for λ for each successive iteration. Second, for exceedingly nonlinear cases, the observable $F(\eta)$ can grow quadratically with respect to the input η , meaning a potentially ill-posed or unbounded inner problem. To resolve these issues, we use the penalty method and add a penalty term that dominates both the action and the Lagrangian term, so that the objective function will truly tend toward the minimum with each iteration:

$$\frac{1}{2}\|\eta\|_{L^2}^2 - \lambda(F(\eta) - z) + \frac{\mu}{2}(F(\eta) - z)^2 \quad (8)$$

which we minimize by parameterizing on η . We update the Lagrange multiplier after each iteration by the following formula,

$$\lambda_{k+1} = \lambda_k + \mu_k(z - F(\eta)) \quad (9)$$

As is the case for solving all optimization problems, the first step is to take the gradient of the objective function (8). We use the `scipy.optimize` package to perform the L-BFGS minimization method, using automatic differentiation which is provided by JAX. The internal mechanisms of the automatic differentiation calls compute the same backward sweep that appears in the Chernykh-Stepanov iterative method described in [2]. After the saddle point approximation, labeling the objective function above as the probability that F reaches the value z is given to leading exponential order by

$$P(F(\eta_z) \geq z) \asymp e^{-\frac{1}{2}\|\eta_z\|^2} \quad (10)$$

Note that the $\asymp$ symbol means equivalence in magnitude, as in $\frac{\log(P(F(\eta_z) \geq z))}{-\frac{1}{2}\|\eta_z\|^2} \rightarrow 1$ as $z \rightarrow \infty$. So we can also claim from (10) that

$$\log(P(F(\eta_z) \geq z)) \approx -\frac{1}{2}\|\eta_z\|^2 \quad (11)$$

We see in Figure 4 how the rate function varies with respect to z , following our expectations laid out above. Note especially the normalization that the PDF follows as the nonlinear coupling weakens (i.e. as $c_1 \rightarrow 0$).

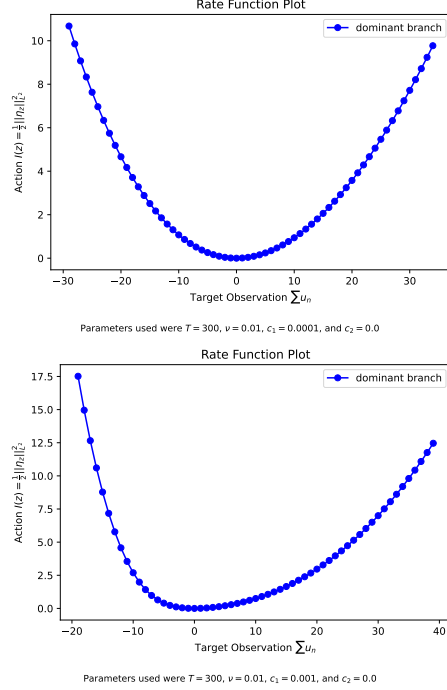
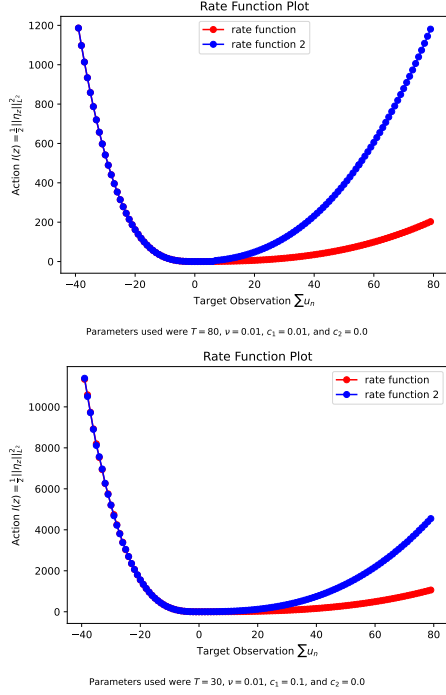


Fig. 4: Rate functions for the target observable of velocity $\sum_n u_n$

Interestingly, we also found that multiple branches of local minima would form at higher levels of nonlinearity, taking shape within the rate function. Though the initial objective function $\frac{1}{2}\|\eta\|_{L^2}^2$ is strictly convex, which would lead us to expect only one local and global minimum, the constraint $F(\eta) = z$ introduces a hypersurface that can be very far from a plane, which would be the only kind of convex set under equality constraints. On this hypersurface, multiple local minima can exist, so it becomes more important to start within "neighborhoods" of related local minima for different z values. Random initializations for η would jump across the branches, so we used a sequential solver

for collections of target observable z values, in order to create a "warm start" for each new optimization.



Still intuitively viewing the action as the "cost" of randomness, we take the branch with the lower action to treat as the dominant branch, which then governs the leading-exponential PDF asymptotics via $\rho(z) \asymp e^{-I(z)}$. More mathematically, because our initial constrained objective function is $\min_{\eta} \frac{1}{2} \|\eta\|_{L^2}^2$, we would clearly choose the branch with the lower action.

Generally, this branch divergence is an expected phenomenon as well from a physical perspective, and in fact we would expect more and more branches to continue appearing as we further increased the nonlinear coupling (i.e. by increasing c_1). This is a natural indicator of the onset of chaos in the system. As discussed in [5], the small-noise asymptotics dominated by a single instanton will break down at some point as c_1 is increased. For example, if you take the Lorenz 63 system in the chaotic regime and add small Brownian noise, and you aim to go from below the attractor to above the attractor with an instanton, there is an infinite number of ways

to do so by going around the butterfly attractor repeatedly.

3.3 Primal and dual problems

The optimization problem we have formulated and solved up to this point is the primal form of the instanton problem. The decision variable is the noise realization η , the dynamics

$$\dot{u} = G[u] + \sqrt{\chi} \eta - \nu k^2 u, \quad u(0) = 0, \quad (12)$$

is treated as a hard constraint that we eliminate by substitution (every forward solve of (12) expresses u as a function of η), and the only constraint that remains explicit is the scalar equality $F(\eta) = z$ at the terminal time, which we enforce through the augmented Lagrangian formulation. In contrast, the formulation used in [2] keeps the dynamics as an explicit constraint, introduces a time-dependent Lagrange multiplier $p(t)$ to enforce it, and solves the resulting saddle-point system by the Chernykh-Stepanov iterative method. This is the dual problem. Both formulations are equivalent at convergence, in the sense that they identify the same instanton η_z and the same action $I(z)$; they differ in which variable is the decision variable and which is implicit.

To make the correspondence precise, consider the full Lagrangian that treats both the dynamics and the terminal observable as constraints:

$$\begin{aligned} \mathcal{L}[\eta, u, p, \lambda] = & \frac{1}{2} \|\eta\|_{L^2}^2 \\ & - \int_0^T \langle p, \dot{u} - G[u] + \nu k^2 u - \sqrt{\chi} \eta \rangle dt \\ & - \lambda (F(u(T)) - z). \end{aligned} \quad (13)$$

The momentum field $p(t)$ is the Lagrange multiplier conjugate to the dynamics (12), and λ is the scalar multiplier conjugate to the terminal constraint. Setting variations of $\mathcal{L}$ to zero yields the Karush-Kuhn-Tucker (KKT) conditions for the instanton:

$$\begin{aligned}
\frac{\delta \mathcal{L}}{\delta \eta} = 0 &\implies \eta(t) = \sqrt{\chi} p(t), \\
\frac{\delta \mathcal{L}}{\delta u} = 0 &\implies -\dot{p} + \nu k^2 p = (\nabla_u G[u])^\top p, \\
&p(T) = \lambda \nabla_u F(u(T)), \\
\frac{\delta \mathcal{L}}{\delta p} = 0 &\implies \dot{u} = G[u] + \chi p - \nu k^2 u.
\end{aligned} \tag{14}$$

The first condition is the algebraic identification of the noise with the momentum at the saddle point: the optimal η is proportional to p through the forcing covariance. The second is the backward adjoint equation that p satisfies, derived by integration by parts on the dynamics term in (13), with a boundary condition at $t = T$ coming from the variation of the observable. The third recovers the forward dynamics (12) with η already eliminated by the first condition. Together, the second and third equations are the coupled instanton equations of [2], and the boundary condition $p(T) = \lambda \nabla_u F(u(T))$ is what the Chernykh-Stepanov scheme uses to initialize its backward sweep.

The two solution strategies that arise from (13) differ in which variable is kept and which is eliminated. In the dual formulation followed by [2], the noise is eliminated via $\eta = \sqrt{\chi} p$ and the algorithm iterates on the coupled (u, p) system: at each step a forward sweep solves for u with p frozen, then a backward sweep solves for p with u frozen. This is required because the linear viscous term has opposite signs in the two equations, $-\nu k^2$ in the forward u equation and $+\nu k^2$ in the backward p equation, and the two are therefore stable only when integrated in opposite time directions. The iteration resolves the coupling by alternation and converges as a fixed-point map. In our primal formulation, we instead keep η explicit and let p be implicit: it never appears as a stored field. Every L-BFGS gradient evaluation runs the forward time-stepper once to produce $u(t; \eta)$, then a reverse-mode automatic differentiation pass walks backward through the time-stepper, producing exactly the discrete adjoint p on the fly. The mechanics of solving the dual problem are discussed in greater detail in [2].

There are tradeoffs to solving the optimization problem using either the primal problem or the dual problem. First, strong duality would

imply that the Lagrange multiplier λ directly corresponds to a target observable z in a bijective manner, which relationship would allow us to graph the rate function directly. However, for strong duality to occur, we would require that the objective function is strictly convex. Of course, in graphing the PDFs of the velocity, velocity gradient, and energy dissipation rate from the direct numerical simulations, we have compelling reason to believe that this is the case, though no formal proof for this claim. If we had only weak duality, then that would introduce the possibility of a duality gap. In the case that the duality gap is finite, the rate function given by the dual problem would be linear. If there was instead an infinite duality gap, then the dual minimization problem would not converge, and the dual problem would not determine an optimal η_z while the primal problem would still correctly find the instanton solution. However, notice that the dual problem is more efficient to solve because for each input value (regardless of if the input is the target observable z or the Lagrange multiplier λ), only one problem needs to be solved. For the primal problem, in contrast, we loop over various values of penalties μ to solve for $F(\eta) = z$, so there are multiple optimization problems being solved when choosing to solve the primal problem. Obviously, this means the computation will be significantly slower for the primal problem. To be thorough and careful about correctness, we opted to solve the primal problem.

3.4 Instanton results

We next inspect the results of the instanton computations. Because Goedert and Biferale solve the dual problem, they parameterize with respect to the Lagrange multiplier λ while we proceed by parameterizing with respect to the target observable z in order to fill in the PDF for each of our three metrics of interest.

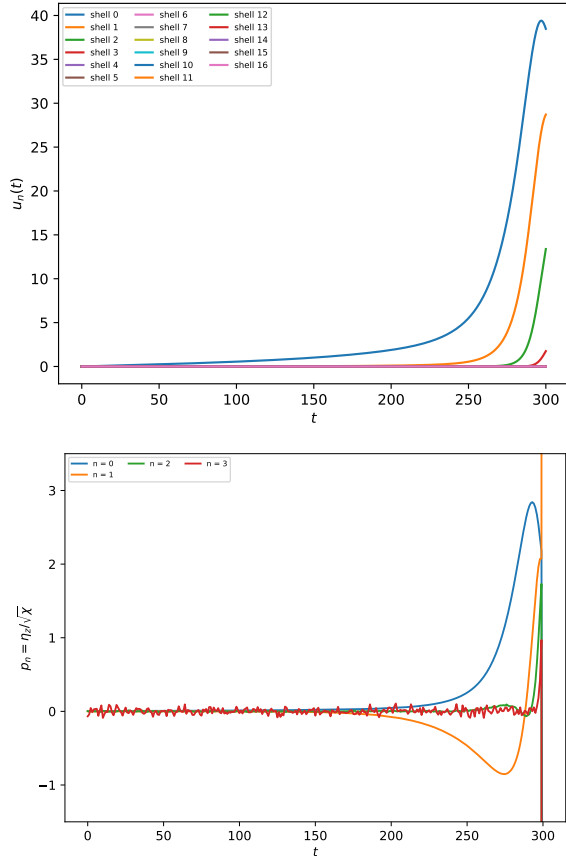


Fig. 6: The plots show the instanton solution for (12), adopting the same definition of p_n as is used in [2] for the plot on the bottom. The parameters used for these plots are $z = \sum_n (u_n) = 82.34$ (corresponding to $\lambda = 2$ as in [2] Figure 5), $c_1 = 10^{-3}$, $c_2 = 0.0$, $\nu = 10^{-2}$, and $\chi_{nm} = k_n^{-6} \delta_{nm}$.

The physical impact of varying the nonlinear coupling c_1 is most clearly seen in the instanton solutions themselves. We fix the target observable value at $z = \sum_n u_n = 3$, hold the remaining parameters constant at $c_2 = 0.0$, $\nu = 0.01$, and $\chi_{nm} = k_n^{-6} \delta_{nm}$, and vary c_1 across three orders of magnitude. The terminal time T is adjusted between runs for efficient convergence at each c_1 , but this does not impact the qualitative shape of the instanton or the shell activation pattern, which are the key features of interest.

The coefficient c_1 controls the strength of the interactions $k_n u_{n-1}^2$ and $k_{n+1} u_n u_{n+1}$ in $G_n[u]$, which are the channels by which energy moves

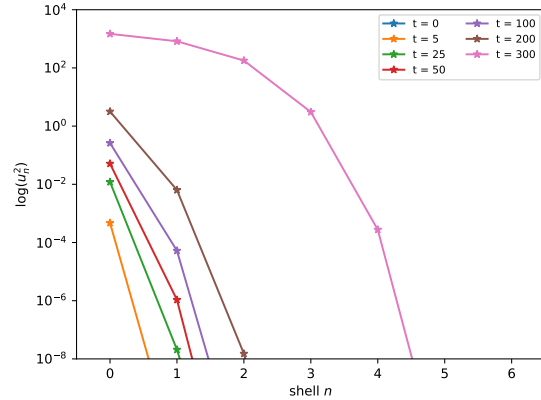


Fig. 7: Squared velocities of each shell in semi-log scale for an instanton approximation as it develops over different time slices, resulting from the parameters $z = \sum_n (u_n) = 82.34$, $c_1 = 10^{-3}$, $c_2 = 0.0$, $\nu = 10^{-2}$, and $\chi_{nm} = k_n^{-6} \delta_{nm}$.

between neighboring shells. When c_1 is small, these channels are nearly closed: energy injected into the system through the stochastic forcing has no efficient route to propagate to other shells before it is dissipated, and the system behaves approximately as a collection of independent Ornstein-Uhlenbeck processes [2]. When c_1 is large, by contrast, the nonlinear coupling dominates, and energy that enters at any one shell can rapidly cascade through the rest of the system.

This contrast is visible directly in Figure 8. At $c_1 = 10^{-4}$ (top), the instanton is concentrated almost entirely in the first shell ($n = 0$). At $c_1 = 10^{-2}$ (middle), several shells contribute non-trivially, and the instanton begins to resemble the multi-scale structure characteristic of a turbulent cascade. At $c_1 = 1$ (bottom), a clear forward-cascade pattern emerges in the time evolution as energy injected at large scales (low n) propagates to higher n before the terminal time.

This progression also illustrates why the saddle-point approximation underlying the instanton method becomes increasingly delicate as c_1 grows. At small c_1 , the dynamics are nearly linear and the action $\frac{1}{2} \|\eta\|_{L^2}^2$ is essentially quadratic in z , giving a single well-defined minimizer and an excellent match between the instanton-predicted PDF and direct numerical simulation. At larger c_1 , the nonlinear coupling enables the multiple-branch structure and

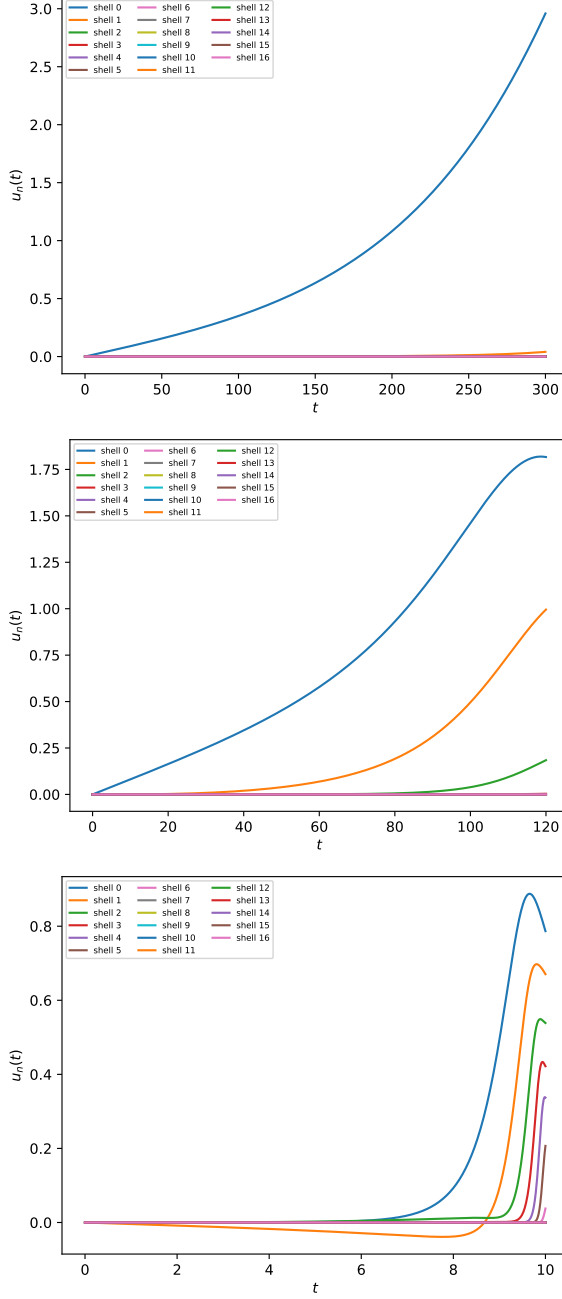


Fig. 8: Instanton solutions for $z = \sum_n u_n = 3$ at varying values of the nonlinear coupling: $c_1 = 10^{-4}$ (top), $c_1 = 10^{-2}$ (middle), and $c_1 = 1$ (bottom). All other parameters held fixed.

the onset of chaotic behavior in the underlying dynamics. The instanton method continues to identify the most likely path for any given

target, but the assumption that a single saddle point dominates the path integral begins to break down, and second-order corrections in the form of a prefactor become increasingly beneficial.

4 Prefactor calculation

In order to make the approximation more accurate, particularly in its shape, we consider second-order fluctuations around the instanton approximation, formed as a prefactor which we denote $c(z)$.

4.1 Prefactor-adjusted PDF derivation

Recall that in the instanton methodology, we define a function F that maps from the noise space to the target observable, i.e. $F(\eta) = z$ for some time stepping noise vector η and some target observable z . We initially approximated its PDF by taking only the leading exponential order of the path integral around F , which resulted in our initial optimization function

$$\min_{\eta} \frac{1}{2} \|\eta\|_{L^2}^2 \quad (15)$$

To derive the prefactor, we simply keep the approximation in greater detail. We define $\eta = \eta_z + \delta\eta$ where η_z is the instanton solution, and expand the constrained action to second order in $\delta\eta$, which gives us the resulting

$$\frac{1}{2} \|\eta_z + \delta\eta\|^2 = \frac{1}{2} \|\eta_z\|^2 + \langle \eta_z, \delta\eta \rangle + \frac{1}{2} \|\delta\eta\|^2 \quad (16)$$

where $I(z) = \frac{1}{2} \|\eta_z\|^2$ already. For the constraint, we want to find $F(\eta) = z$ while taking a second-order approximation of F , and since the first term in the Taylor expansion is z itself, we are left with

$$\langle \nabla_{\eta} F, \delta\eta \rangle + \frac{1}{2} \langle \delta\eta, \nabla_{\eta}^2 F \cdot \delta\eta \rangle = 0 \quad (17)$$

By combining the action and the constraint, we are left with

$$\delta(F - z) = \int \frac{d\lambda}{2\pi} e^{i\lambda(F-z)} \quad (18)$$

From here, we approximate the resulting Gaussian integral. Note that the standard Gaussian integral in finite dimensions is

$$\int e^{-\frac{1}{2}x^T A x} dx = \sqrt{\frac{(2\pi)^n}{\det(A)}} \quad (19)$$

where the determinant is taken on the tangent space to the constraint surface. We extend this formula analogously to infinite dimensions, noting that the constant factors will be normalized to be a PDF, resulting in

$$\rho(z) \approx \frac{1}{\sqrt{2\pi}} \cdot \frac{|\lambda_z|}{\sqrt{\det(A)}} e^{-I(z)} \quad (20)$$

This is our core PDF approximation with the prefactor adjustment, meaning that the prefactor is left to be

$$\begin{aligned} c(z) &= \frac{|\lambda_z|}{\sqrt{\det(A)}} \\ &= \frac{|\lambda_z|}{\sqrt{2I(z) \cdot \det(I - B)}} \end{aligned} \quad (21)$$

where A is an infinite-dimensional matrix and $I - B$ provides a truncated, finite-dimensional approximation of A .

4.2 Computing the prefactor

In the prefactor equation (21), λ_z and $I(z)$ are both readily available from the initial instanton computation. The only marginal computation for the prefactor is determining the value of $\det(I - B)$, where $B_z = \lambda_z \nabla_\eta^2 F$ evaluated at η_z . However, most Hessians are already time-consuming and challenging to solve efficiently, and because the size of this Hessian is dependent on the number of time steps and shells being used, effectively reaching an order in the tens of thousands, the standard way of finding the determinant of B_z is not computationally feasible.

Instead, pulling from [4], we approximate the Hessian by collecting eigenvalues using a matrix-free iterative eigensolver (ARPACK, using the Arnoldi method), then calculating

$$\det(I - B_z) \approx \prod_k^K (1 - \mu_k) \quad (22)$$

for a sufficiently large number K such that the product converges. We also know that the eigenvalues μ_k will tend toward 0 because B_z is a compact operator on the noise space. This is because B_z is built out of the forward map $F(\eta) = z$ which has smoothing properties as a result of the viscous dissipation implicit in the physical system.

Looking at the cumulative products $\prod_k^K (1 - \mu_k)$ in the insets of the plots in Figure 9, it is clear that this product converges towards a finite determinant despite being the truncation of an infinite-dimensional system. The number of eigenvalues varies with the target metric being measured and the other parameters, such as the target observable value z . Note that as z goes further from 0, the number of eigenvalues required for the product to converge get much larger. Also, as c_1 increases, the number of eigenvalues required for the product to converge also grow larger. Both results are intuitive; larger z pushes the instanton into more strongly nonlinear regions of phase space, and larger c_1 values strengthen the nonlinear coupling between shells. In both cases, nonlinear amplification competes with viscous dissipation, so the second variation B_z becomes less smoothing, leading to the eigenvalues decaying more slowly.

With this result complete, we can continue by inspecting the resultant PDFs of the two methodologies in comparison to the direct numerical simulations of the D.N. shell model.

4.3 Accuracy deterioration analysis

We can begin by inspecting the comparison of excess kurtosis for the velocity and velocity gradient metrics, both of which are linear metrics. We use the following formula to determine the excess kurtosis for a given c_1 value:

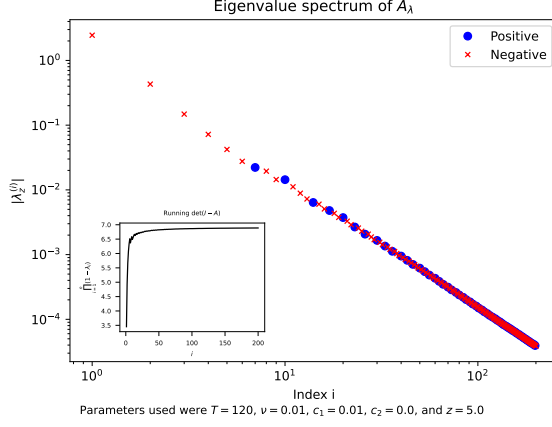


Fig. 9: Eigenvalue spectra with velocity as the target observable metric

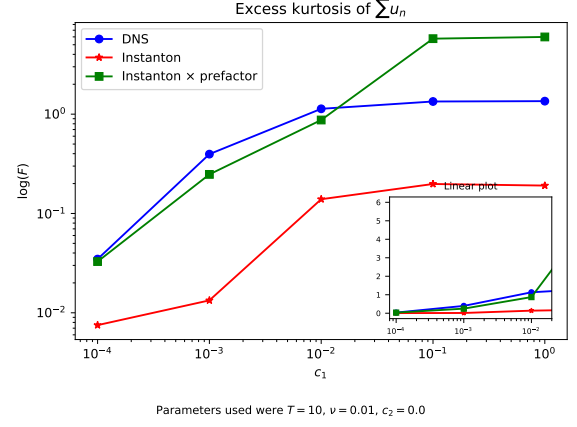
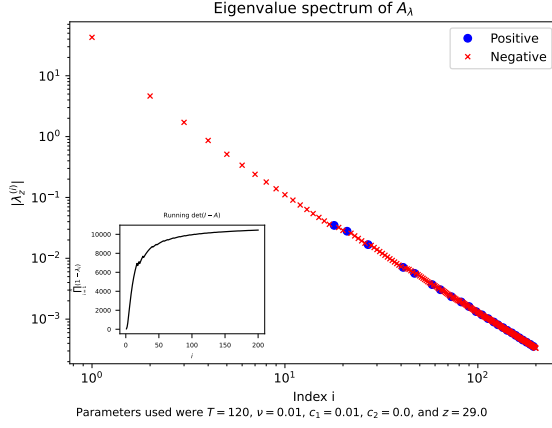


Fig. 10: Excess kurtosis plots for velocity (top) and velocity gradient (bottom) with respect to c_1 .

$$F = \frac{\langle (O - \langle O \rangle)^4 \rangle}{\langle (O - \langle O \rangle)^2 \rangle^2} - 3 \quad (23)$$

The energy dissipation rate is non-negative by construction, and therefore definitely not Gaussian, so it can not be logically assessed by this same excess kurtosis analysis which is based on a Gaussian reference point.

For the kurtosis of the prefactor-adjusted instantons, the result is approximately what we would expect to see. For linear observables, we expect the kurtosis of the instanton computations to track the direct numerical simulation kurtosis for near-Gaussian levels of c_1 , but as nonlinearity grows, we would expect the instanton-computed PDFs to grow much more strongly than the real-world kurtosis. Similarly, for the prefactor-adjusted instanton-computed PDFs, we would expect the kurtosis of these

PDFs to track the direct numerical simulation kurtosis at near-Gaussian parameters as well, and then grow at a much slower and better adjusted pace than the instanton-only PDF.

However, the kurtosis of the instanton PDFs seem to be much better here than expected, significantly slowing in growth, to the point where it is showing less kurtosis than the direct numerical simulation. This is an unexpected result, and it would likely require a closer look to determine why this is occurring. In particular, we note that this is a very different result than is found in [2].

Generally, the trend we see matches the trend that we would expect. There is approximately no excess kurtosis for small values of c_1 , as these PDFs appear to be approximately Gaussian, which leads to more accurate approximations in the instanton computations. As c_1 grows, we expect the kurtosis to deteriorate over

time and drift further and further away from the excess kurtosis predicted by the DNS as the accuracy of instanton-based computations tends to deteriorate as nonlinearity is strengthened, even with a prefactor.

5 Conclusion

We conclude that the Desniansky-Novikov shell model remains an excellent testing ground for instanton-based computations, and that the methodology extends naturally to incorporate prefactor corrections beyond the leading-order saddle-point approximation. The Martin-Siggia-Rose-Janssen-de Dominicis formalism, originally applied to shell models in the leading-exponential regime by [2], was reformulated as a primal-form optimization problem that we then solved. Building on this primal-form solver, we then introduced the second-order correction $c(z)$ following the framework of [3, 4], computing the Fredholm determinant $\det(I - B_z)$ by extracting the dominant eigenvalues of $B_z = \lambda_z \nabla_\eta^2 F|_{\eta_z}$ via matrix-free Arnoldi iteration, with each matrix-vector product realized as a composition of JAX automatic-differentiation calls through the time-stepper.

Starting from the heat-equation limit $c_1 \rightarrow 0$ provides excellent validation ground, since in this regime the model reduces to a sequence of Ornstein-Uhlenbeck processes for which analytic results are available; both the leading-order instanton PDF and the prefactor-corrected PDF recover the expected Gaussian asymptotics. As nonlinearity is increased, we observed good agreement between DNS, leading-order instantons, and prefactor-corrected instantons in the regime where dissipative dynamics dominate, even as the distributions deviate from Gaussianity due to increasing asymmetry. The prefactor correction in particular sharpens the agreement at moderate z , where the instanton approximation alone misses the polynomial-scale normalization of the tail.

However, the prefactor correction does not, on its own, resolve the deeper limitations of the method. As soon as nonlinearity becomes relevant in the first few shells, two qualitatively new phenomena appear that lie outside the scope of a single quadratic expansion around a single instanton. First, the rate function develops multiple branches at higher c_1 , corresponding

to distinct local minima of the constrained action on the same target surface. The prefactor formula remains valid on each branch separately, but the leading-exponential PDF is governed by whichever branch has the lower action at a given z , and tracing the dominant branch globally requires careful warm-starting and branch-resolved continuation. Second, even on the dominant branch, the agreement between instanton-based predictions and DNS deteriorates at large c_1 , with the instanton method predicting heavier positive tails than DNS measures.

We expect that the regime of validity can be extended further by computing higher-order corrections in the functional series expansion of the action — i.e., by going beyond the Gaussian-fluctuation approximation that the prefactor encodes and incorporating cubic and quartic terms in $\delta\eta$. The methods of [4] provide a natural starting point. A complementary direction concerns the choice of observable: we have focused on $\sum_n u_n$, $\sum_n k_n u_n$, and $\nu \sum_n k_n^2 u_n^2$, but other physically motivated observables — particularly those concentrated at small scales — likely have different sensitivities to nonlinearity and may exhibit a wider range of validity. The branch phenomenon itself also warrants further investigation, both as a numerical challenge and as a physical indicator of the onset of chaos in the underlying dynamics, as discussed in [5] for the Lorenz system.

The formalism developed here is not specific to the Desniansky-Novikov model: the primal-form solver and the prefactor pipeline apply to any shell model by changing the coupling $G_n[u]$ and the observable F . Natural extensions include the GOY and SABRA shell models for hydrodynamic turbulence, the complex-valued variants of D.N. with their richer chaotic attractors, and the shell models for magneto-hydrodynamics, passive scalars, and convective turbulence reviewed in [2]. The open scientific question is in each case how far the saddle-point expansion can be pushed before higher-order corrections or non-perturbative effects take over.

6 Acknowledgements

I would like to warmly thank my advisor, Prof. Timo Schorlepp, for his guidance, patience, and

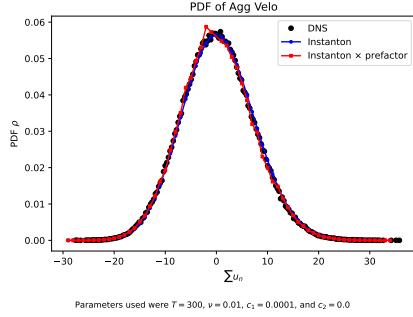
kindness throughout this process. I learned so much not only about fluid dynamics and large deviation theory, but also about the research process and best practices that make research more efficient and effective.

7 Appendix of PDFs

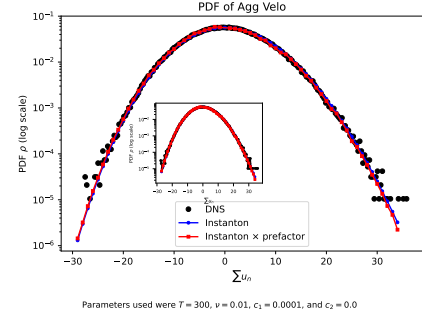
The PDFs, computed in each of the three different methodologies, and for each of the three target observable metrics, are plotted below. We conclude that the prefactor adjustment is beneficial for greater nonlinear couplings than instanton computations alone, but it also has a limited accuracy. In particular, the prefactor adjustments retain the same shape as the DNS for greater values of c_1 than do the instantons alone, which means that we could theoretically determine a multiplicative constant on each of the tails to "shift" the PDFs closer to those presented by the DNS.

The prefactor-adjusted instanton-computed PDFs should, however, be smooth curves, which clearly is not the case in some of the more nonlinear settings. This is another issue that would require more time to properly inspect and resolve.

7.1 Velocity

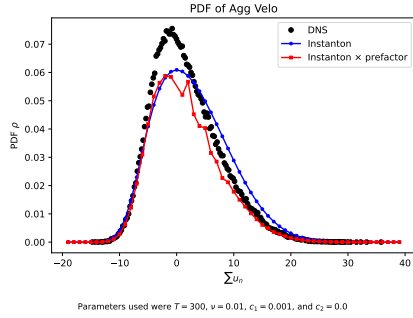


(a) Linear.

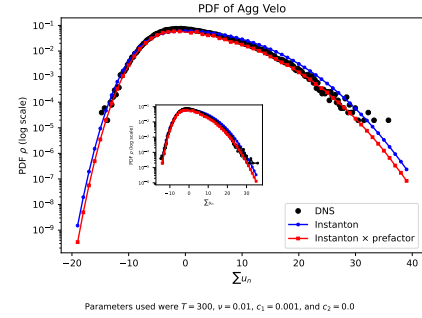


(b) Semilog.

Fig. 11: Velocity PDF, $c_1 = 10^{-4}$.

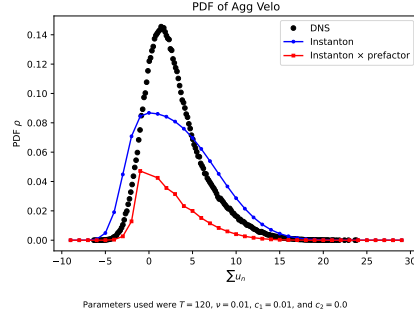


(a) Linear.

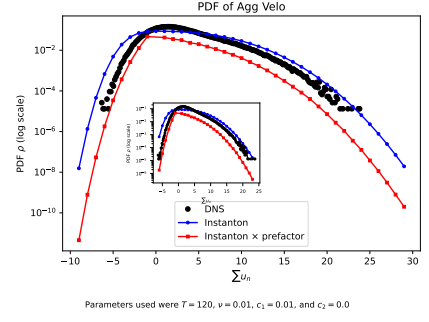


(b) Semilog.

Fig. 12: Velocity PDF, $c_1 = 10^{-3}$.

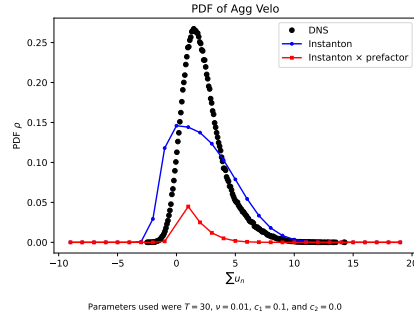


(a) Linear.

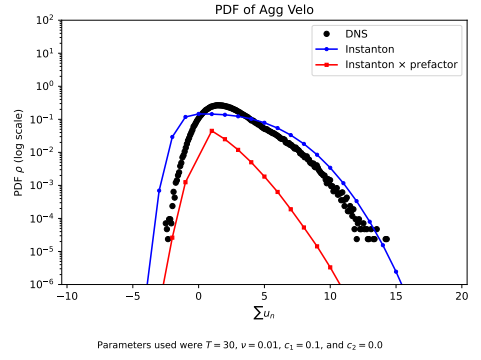


(b) Semilog.

Fig. 13: Velocity PDF, $c_1 = 10^{-2}$.

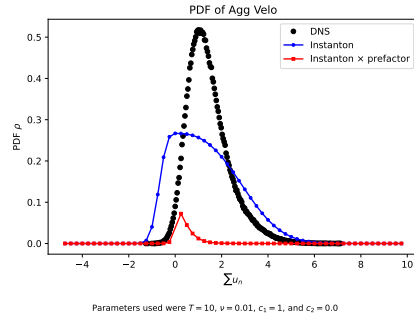


(a) Linear.

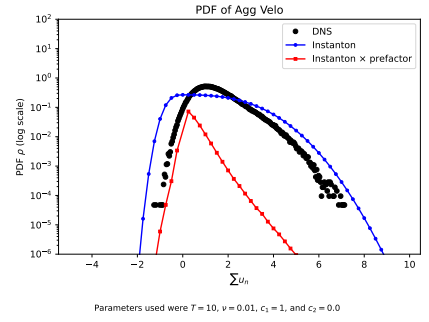


(b) Semilog.

Fig. 14: Velocity PDF, $c_1 = 10^{-1}$.



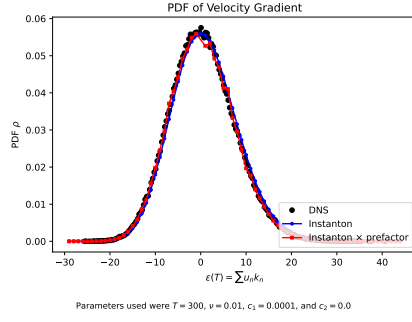
(a) Linear.



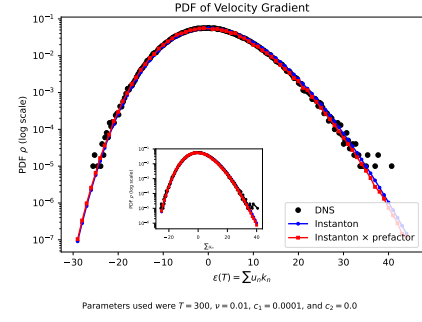
(b) Semilog.

Fig. 15: Velocity PDF, $c_1 = 10^0$.

7.2 Velocity Gradient

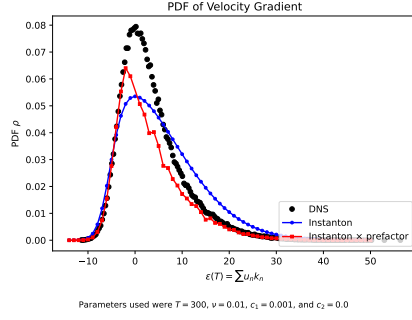


(a) Linear.

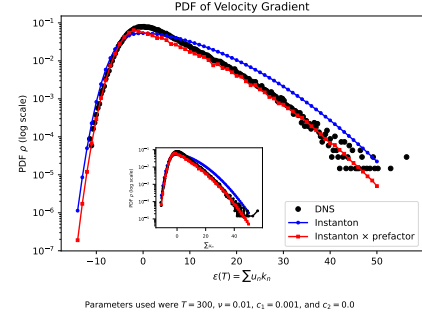


(b) Semilog.

Fig. 16: Velocity Gradient PDF, $c_1 = 10^{-4}$.

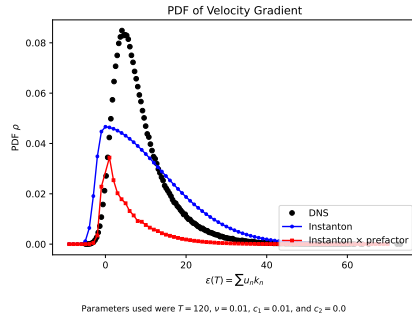


(a) Linear.

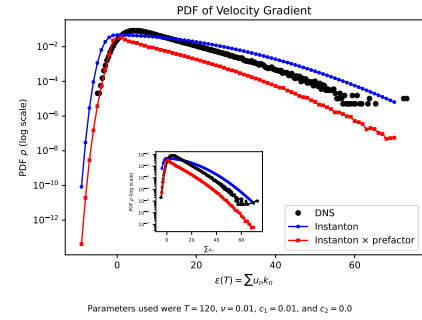


(b) Semilog.

Fig. 17: Velocity Gradient PDF, $c_1 = 10^{-3}$.

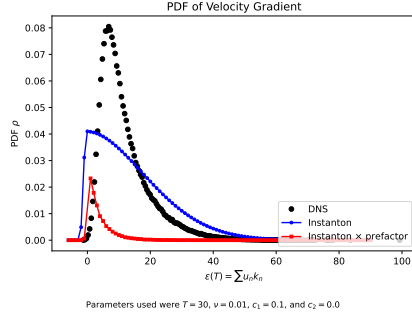


(a) Linear.

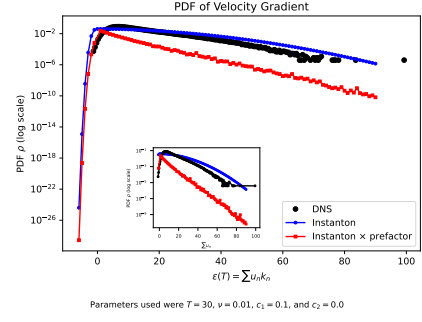


(b) Semilog.

Fig. 18: Velocity Gradient PDF, $c_1 = 10^{-2}$.

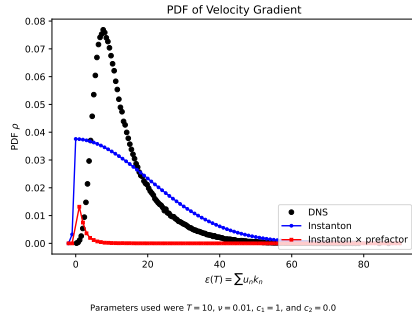


(a) Linear.

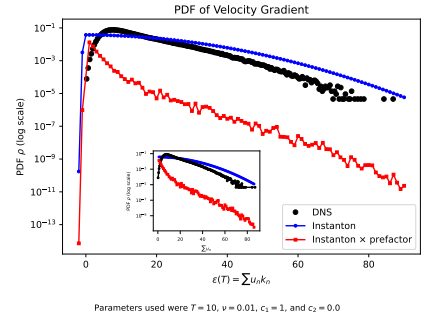


(b) Semilog.

Fig. 19: Velocity Gradient PDF, $c_1 = 10^{-1}$.



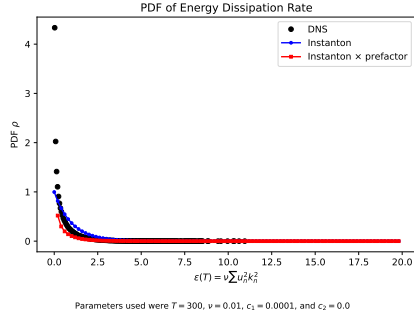
(a) Linear.



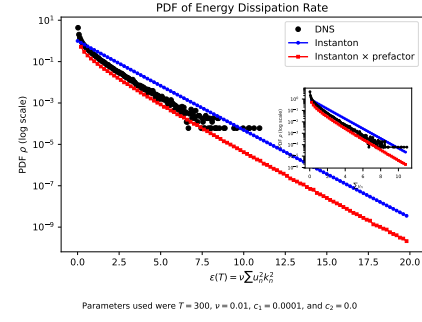
(b) Semilog.

Fig. 20: Velocity Gradient PDF, $c_1 = 10^0$.

7.3 Energy Dissipation Rate

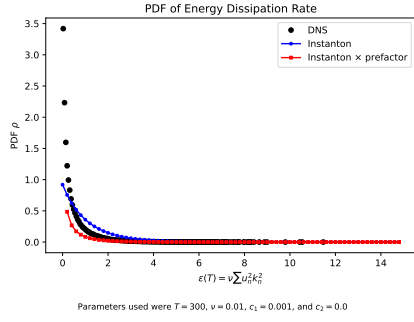


(a) Linear.

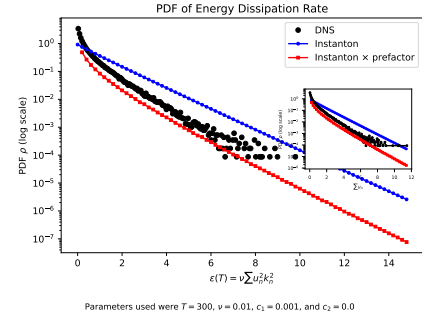


(b) Semilog.

Fig. 21: Energy Dissipation Rate PDF, $c_1 = 10^{-4}$.

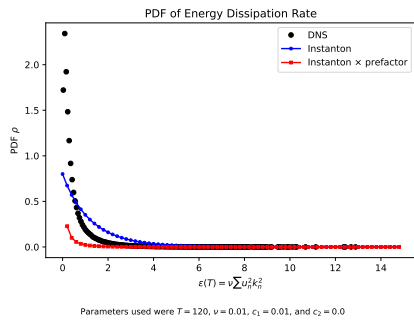


(a) Linear.

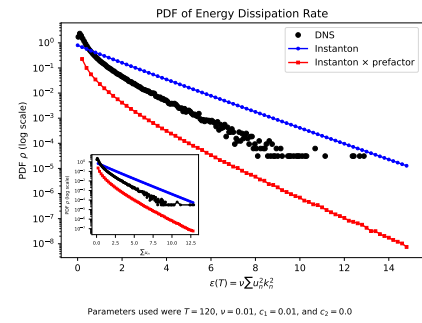


(b) Semilog.

Fig. 22: Energy Dissipation Rate PDF, $c_1 = 10^{-3}$.

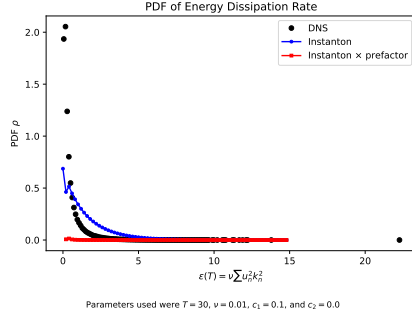


(a) Linear.

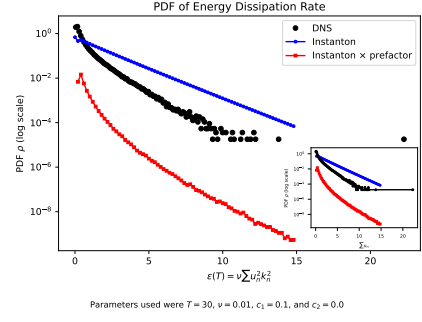


(b) Semilog.

Fig. 23: Energy Dissipation Rate PDF, $c_1 = 10^{-2}$.

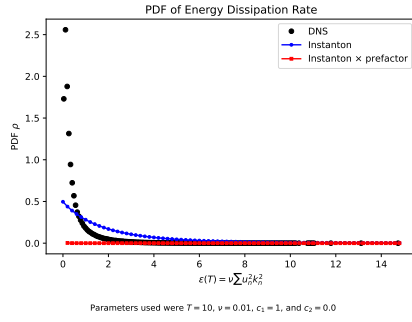


(a) Linear.

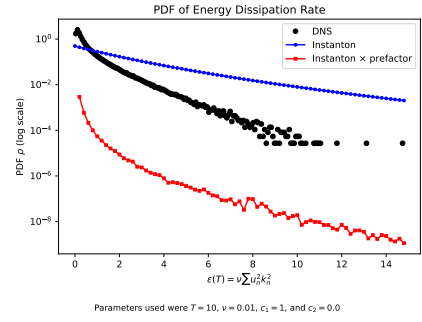


(b) Semilog.

Fig. 24: Energy Dissipation Rate PDF, $c_1 = 10^{-1}$.



(a) Linear.



(b) Semilog.

Fig. 25: Energy Dissipation Rate PDF, $c_1 = 10^0$.

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